

# ADVANCED COMPUTER SCIENCE, INFORMATICS & ARTIFICIAL INTELLIGENCE

New Egyptian Baccalaureate Official Framework - Grade 12

ACADEMIC YEAR: 2025 / 2026 | EG-BAC GRADE 12 (ADVANCED COMPUTATIONAL & EMERGING TECHNOLOGIES DIVISION)

## CURRICULUM OVERVIEW & AUTHORIZED SCOPE

Rigorous collegiate-level computer science, systems engineering, and machine learning framework designed for the New Egyptian Baccalaureate, incorporating hardware design, balanced tree algorithms, database normalization, zero-trust network defense, and neural network deep learning pipelines.

### CORE CURRICULUM HIGHLIGHTS:

- Pipelined CPU architecture, superscalar execution, and cache coherence protocols
- Self-balancing AVL trees, red-black tree invariants, and cryptographic hash maps
- Dynamic programming recurrences, Master Theorem asymptotics, and divide-and-conquer bounds
- Python metaclasses, abstract base classes, decorators, and asynchronous concurrency
- Advanced SQL window functions, transaction isolation levels, and ACID deadlock resolution
- IPv4/IPv6 dual-stack routing, BGP/OSPF convergence, and TCP congestion control algorithms
- Zero-trust network architecture, public key infrastructure (PKI), and penetration defense
- Multi-layer perceptron gradient descent backpropagation, loss surfaces, and explainable AI

### APPROVED CHAPTER MODULES:

|                                                                      |                             |
|----------------------------------------------------------------------|-----------------------------|
| - Unit 1: Digital Logic, Hardware Architecture & Processor Internals | pp. 1-27 (15 core units)    |
| - Unit 2: Data Representation, Number Systems & Memory Organization  | pp. 28-54 (15 core units)   |
| - Unit 3: Linear & Non-Linear Data Structures: Stacks, Queues, Trees | pp. 55-82 (15 core units)   |
| - Unit 4: Algorithms, Pseudocode & Computational Complexity          | pp. 83-110 (15 core units)  |
| - Unit 5: Structured & Object-Oriented Programming with Python       | pp. 111-137 (15 core units) |
| - Unit 6: Relational Databases, Normalization & Advanced SQL         | pp. 138-165 (15 core units) |
| - Unit 7: Full-Stack Web Technologies & Client-Server Architecture   | pp. 166-192 (15 core units) |
| - Unit 8: Computer Networking, OSI/TCP-IP Protocols & Routing        | pp. 193-220 (15 core units) |
| - Unit 9: Cybersecurity, Cryptography & Threat Prevention            | pp. 221-247 (15 core units) |

## OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: EG-Bac Advanced Computational Systems, Algorithms & AI Compendium (52p - 5.2 MB)

Full Ministerial Textbook: 275 Pages - 31.2 MB (Direct on MOE Portal)

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# MINISTERIAL PREFACE & STATUTORY FRAMEWORK

## MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

### 1. Pedagogical Scope & National Objectives

Rigorous collegiate-level computer science, systems engineering, and machine learning framework designed for the New Egyptian Baccalaureate, incorporating hardware design, balanced tree algorithms, database normalization, zero-trust network defense, and neural network deep learning pipelines. Pipelined CPU architecture, superscalar execution, and cache coherence protocols Self-balancing AVL trees, red-black tree invariants, and cryptographic hash maps Dynamic programming recurrences, Master Theorem asymptotics, and divide-and-conquer bounds Python metaclasses, abstract base classes, decorators, and asynchronous concurrency Advanced SQL window functions, transaction isolation levels, and ACID deadlock resolution IPv4/IPv6 dual-stack routing, BGP/OSPF convergence, and TCP congestion control algorithms Zero-trust network architecture, public key infrastructure (PKI), and penetration defense Multi-layer perceptron gradient descent backpropagation, loss surfaces, and explainable AI

### 2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

|                                                                                              |                                                                                                        |                                                                                                         |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <div>30%</div> <div>Knowledge &amp; Recall</div> <div>Definitions, laws &amp; formulas</div> | <div>45%</div> <div>Understanding &amp; Application</div> <div>Calculations, proofs &amp; models</div> | <div>25%</div> <div>Higher Order Thinking (HOTS)</div> <div>Analysis, synthesis &amp; non-routine</div> |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|

### 3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

### 4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

# SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

| Chapter / Modular Title                                             | Weeks | Marks | Weight | Curricular Focus                                            |
|---------------------------------------------------------------------|-------|-------|--------|-------------------------------------------------------------|
| Unit 1: Digital Logic, Hardware Architecture & Processor Internals  | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 2: Data Representation, Number Systems & Memory Organization   | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 3: Linear & Non-Linear Data Structures: Stacks, Queues, Trees  | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 4: Algorithms, Pseudocode & Computational Complexity           | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 5: Structured & Object-Oriented Programming with Python        | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 6: Relational Databases, Normalization & Advanced SQL          | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 7: Full-Stack Web Technologies & Client-Server Architecture    | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 8: Computer Networking, OSI/TCP-IP Protocols & Routing         | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 9: Cybersecurity, Cryptography & Threat Prevention             | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |
| Unit 10: Artificial Intelligence, Machine Learning & Digital Ethics | 3 wks | 6 pts | 10.0%  | 15 core instructional units covering national standards and |

## INTERDISCIPLINARY COGNITIVE BRIDGES

|                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Physics (Mechanics &amp; Electromagnetism)</b><br>Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators. |
| <b>Engineering Sciences (Civil, Mechanical, Electrical)</b><br>Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.               |
| <b>Computer Science &amp; Algorithmics</b><br>Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.      |

CHAPTER SPECIFICATION: UNIT 1: DIGITAL LOGIC, HARDWARE ARCHITECTURE & PROCESSOR INTERNALS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 1: Digital Logic, Hardware Architecture & Processor Internals.
- [+] Synthesize and apply analytical frameworks across all 15 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Unit 1: Digital Logic, Hardware Architecture & Processor Internals.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric:  $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification:  $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model:  $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: UNIT 2: DATA REPRESENTATION, NUMBER SYSTEMS & MEMORY ORGANIZATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 2: Data Representation, Number Systems & Memory Organization.
- [+] Synthesize and apply analytical frameworks across all 15 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Unit 2: Data Representation, Number Systems & Memory Organization.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric:  $\Phi_2 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification:  $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model:  $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

## CHAPTER SPECIFICATION: UNIT 3: LINEAR & NON-LINEAR DATA STRUCTURES: STACKS, QUEUES, TREES

### A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 3: Linear & Non-Linear Data Structures: Stacks, Queues, Trees.
- [+] Synthesize and apply analytical frameworks across all 15 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

### B. Statutory Theorems & Analytical Principles

**Ministerial Framework Standard 3.1:** Primary foundational doctrines and axiomatic structures of Unit 3: Linear & Non-Linear Data Structures: Stacks, Queues, Trees.

**Curricular Analytical Directive 3.2:** Standard methodological paradigms and diagnostic criteria.

### C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric:  $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification:  $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model:  $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

### D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

# OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

| ADVANCED COMPUTER SCIENCE, INFORMATICS & ARTIFICIAL INTELLIGENCE                                    |
|-----------------------------------------------------------------------------------------------------|
| Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$                                  |
| Equilibrium Balance Criterion: $\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$                 |
| Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \frac{\partial \rho}{\partial t} = 0$ |
| Normalized Standard Score: $Z = \frac{(X - \mu)}{\sigma}$                                           |

| PROCEDURAL ASSESSMENT & TRANSFORMATION RULES                                                       |
|----------------------------------------------------------------------------------------------------|
| Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$     |
| Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$                                            |
| Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$                       |
| Curricular Efficiency Ratio: $\eta = (\text{Output\_actual} / \text{Input\_standard}) \cdot 100\%$ |

# NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

## 1. Structure of the National Examination Paper

|                          |                                                                                    |
|--------------------------|------------------------------------------------------------------------------------|
| Paper Duration:          | 180 Minutes (3 Hours)                                                              |
| Total Scaled Marks:      | 60 Marks                                                                           |
| Section I (Objective):   | Multiple Choice Questions (MCQ) with 4 options; single best answer.                |
| Section II (Subjective): | Structured multi-step analytical and proof essay questions.                        |
| Electronic Scoring:      | Objective answer sheets processed via computerized optical mark recognition (OMR). |
| Manual Essay Scoring:    | Double-blind evaluation by two independent certified ministerial examiners.        |

## 2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION  
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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